Phys 153: Fundamentals of Physics III

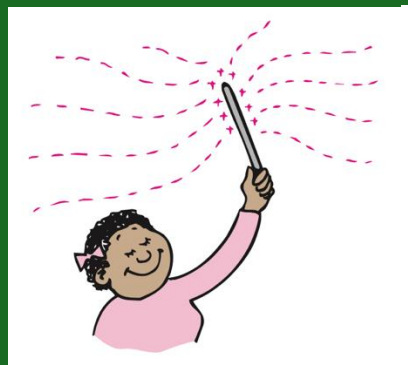
Unit #1 – Ray Optics & Images

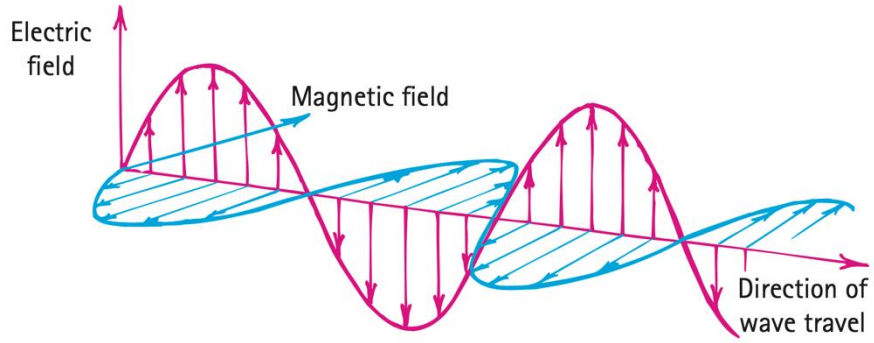
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Spring 2027

Electromagnetic Waves

- By 1887, many aspects of light could be understood by the electromagnetic wave model.
- Key properties of all electromagnetic (EM) waves:
  - EM waves can travel through a vacuum and do so with a definite and unchanging speed $c = 299\,792\,458$ m/s.
  - EM waves are transverse: $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ are perpendicular to each other and to the direction of wave propagation, which is given by $\vec{\mathbf{E}} \times \vec{\mathbf{B}}$ (vector cross product).
  - At any instant, there is a definite ratio between the electric and magnetic field strengths: $|\vec{\mathbf{E}}| = c|\vec{\mathbf{B}}|$



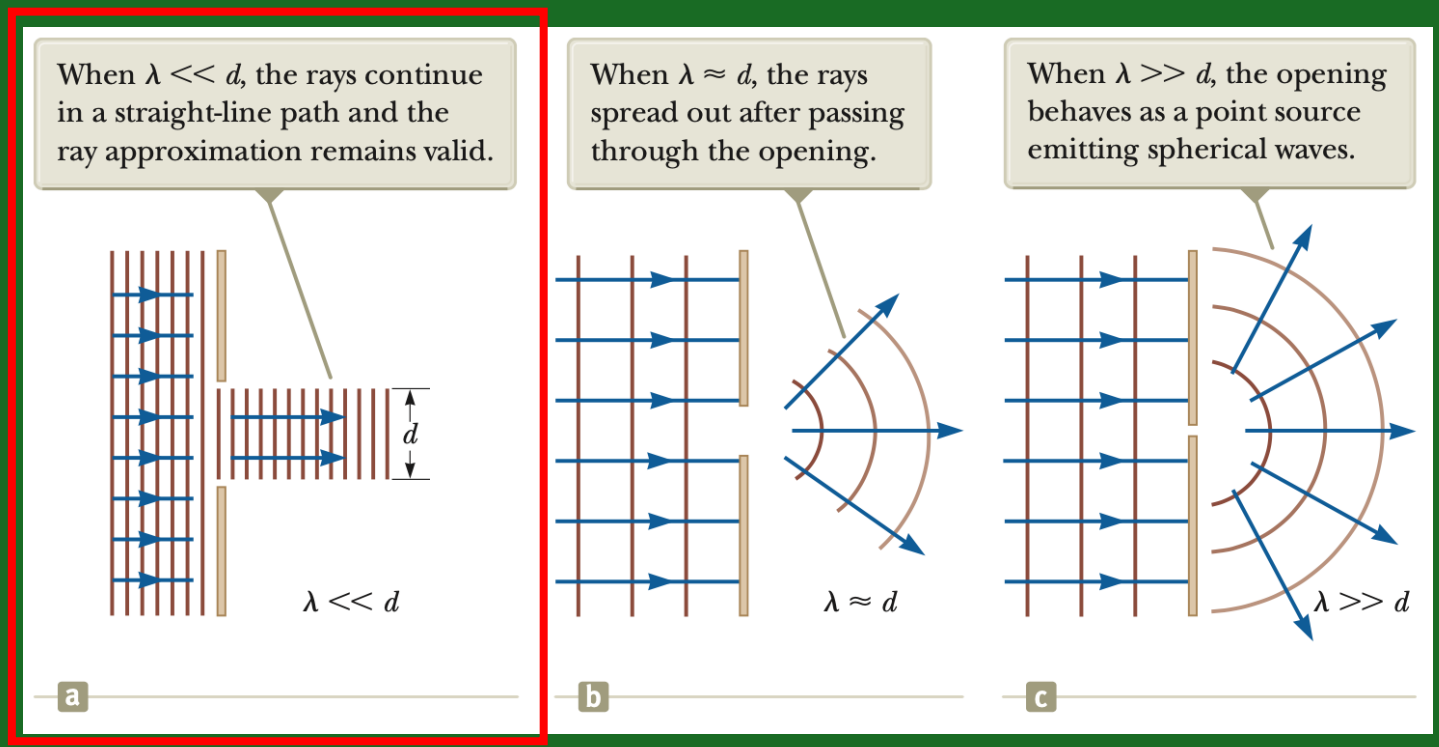


Review of Waves

- A wave is any disturbance of a field that travels through space, carrying energy without necessarily transporting matter.
- The phase of a wave specifies where in its oscillation the wave is at a particular point and time.
  - Points with the same phase oscillate in step; they reach max and min at the same time.
  - Points that are 180° out of phase oscillate out of step; one point is at max while the other is at min.
- A wavefront is a surface connecting all points that share the same phase at a given instant.
  - For a plane wave, the wavefronts are parallel planes.
  - For a spherical wave, the wavefronts are concentric spheres centered on the source.

Ray Approximation

- The rays of a wave are straight lines perpendicular to the wavefronts.
- In the ray approximation, a wave moving through a uniform medium travels in a straight line in the direction of its rays. Valid when the wavelength is “small.”



Principle of Least Time

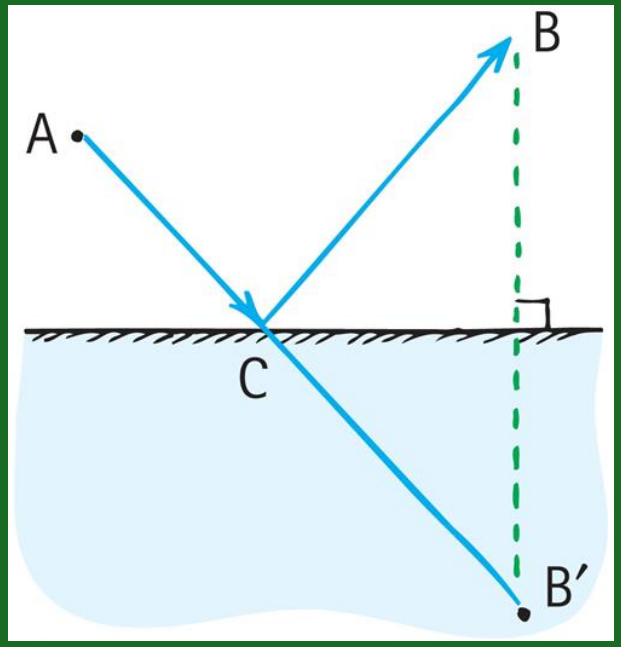
The *principle of least time* states that the path taken by a ray between two given points is the path that can be traveled in the least amount of time.

- Formulated by Pierre de Fermat before it was known that light was an electromagnetic wave.
- We can use the ray model and the principle of least time to explain certain optical phenomena.

Quickest Path for Reflection

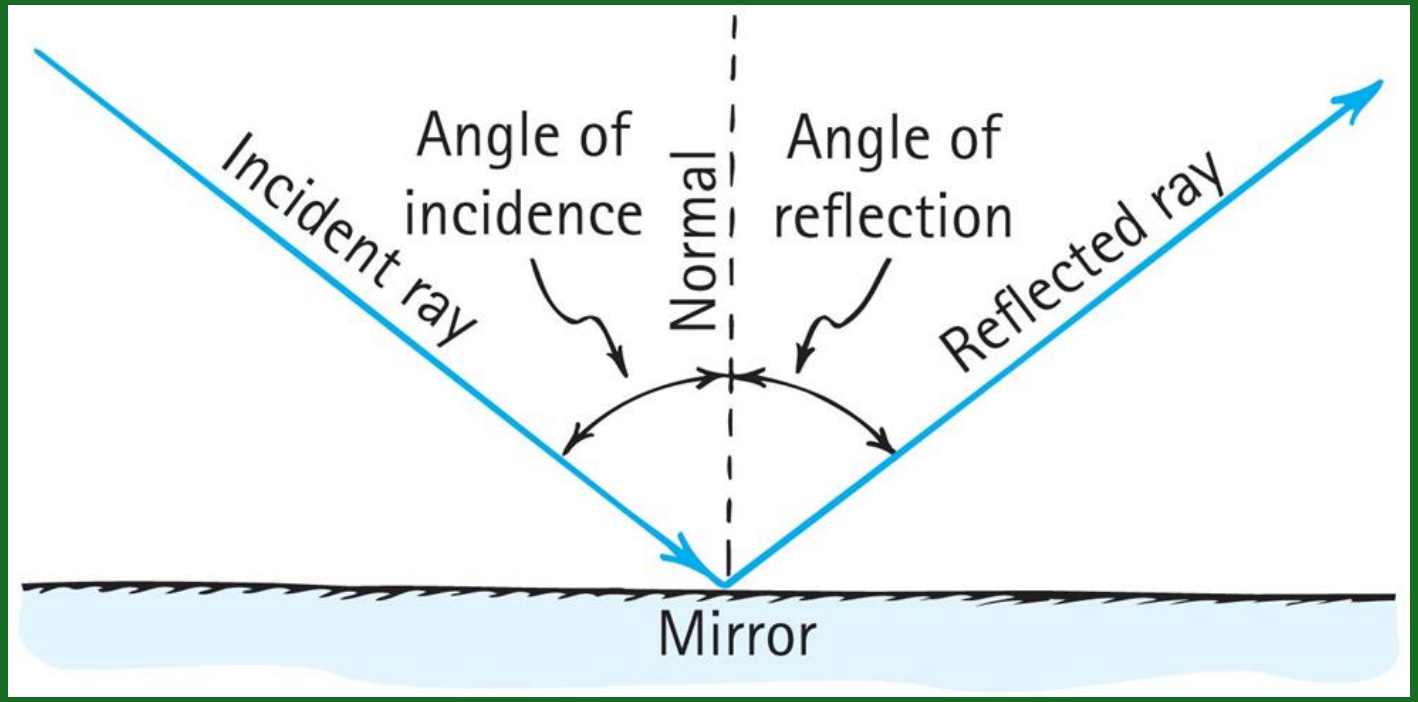
Finding the shortest time for light to go from A to B by reflecting off the mirror:

- Construct, on the opposite side of the mirror, an artificial point B', which is the same distance below the mirror as the point B is above the mirror.
- The shortest distance between A and B' is a straight line and $\overline{AB'} = \overline{AB}$ by symmetry.
- This straight line intersects the mirror at a point C, the precise point of reflection for least time from A to B.



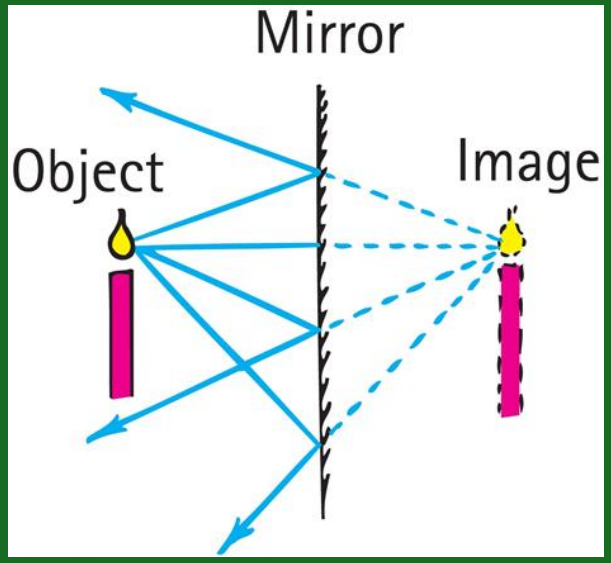
Law of Reflection

The angle of incidence equals the angle of reflection.



Plane Mirrors

- A plane mirror creates a virtual image that is
  - is the same size as the object, formed behind a mirror, and located at the position where the extended reflected rays converge.
  - is as far behind the mirror as the object is in front of the mirror.

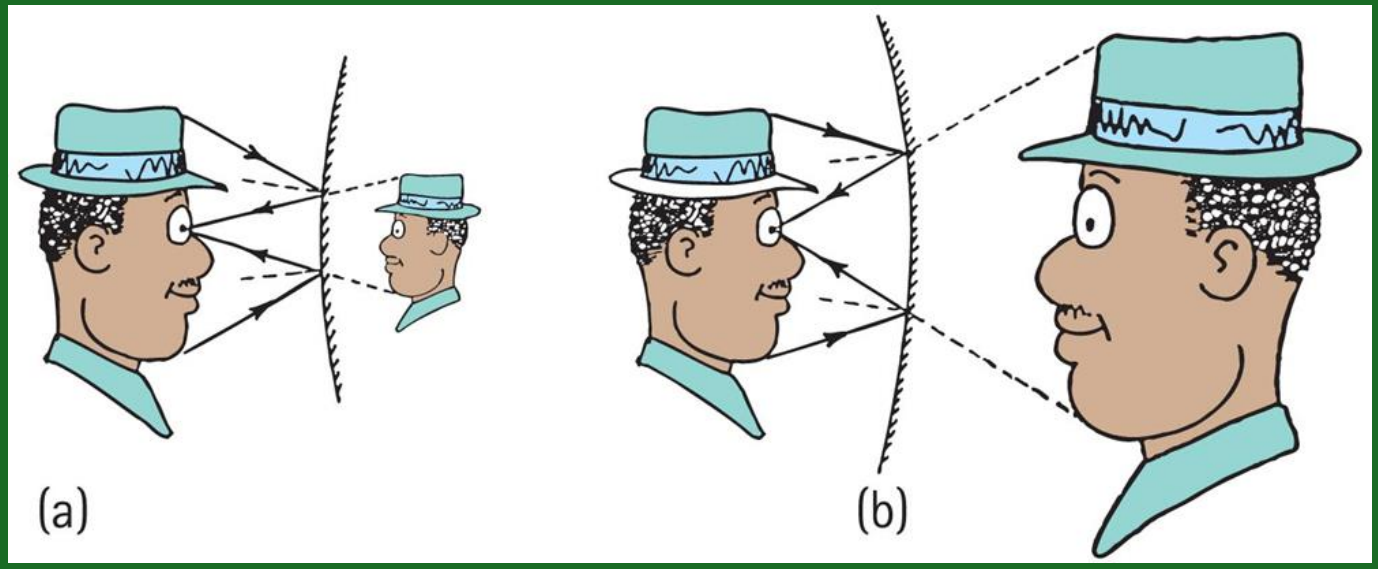


How does the mirror know?

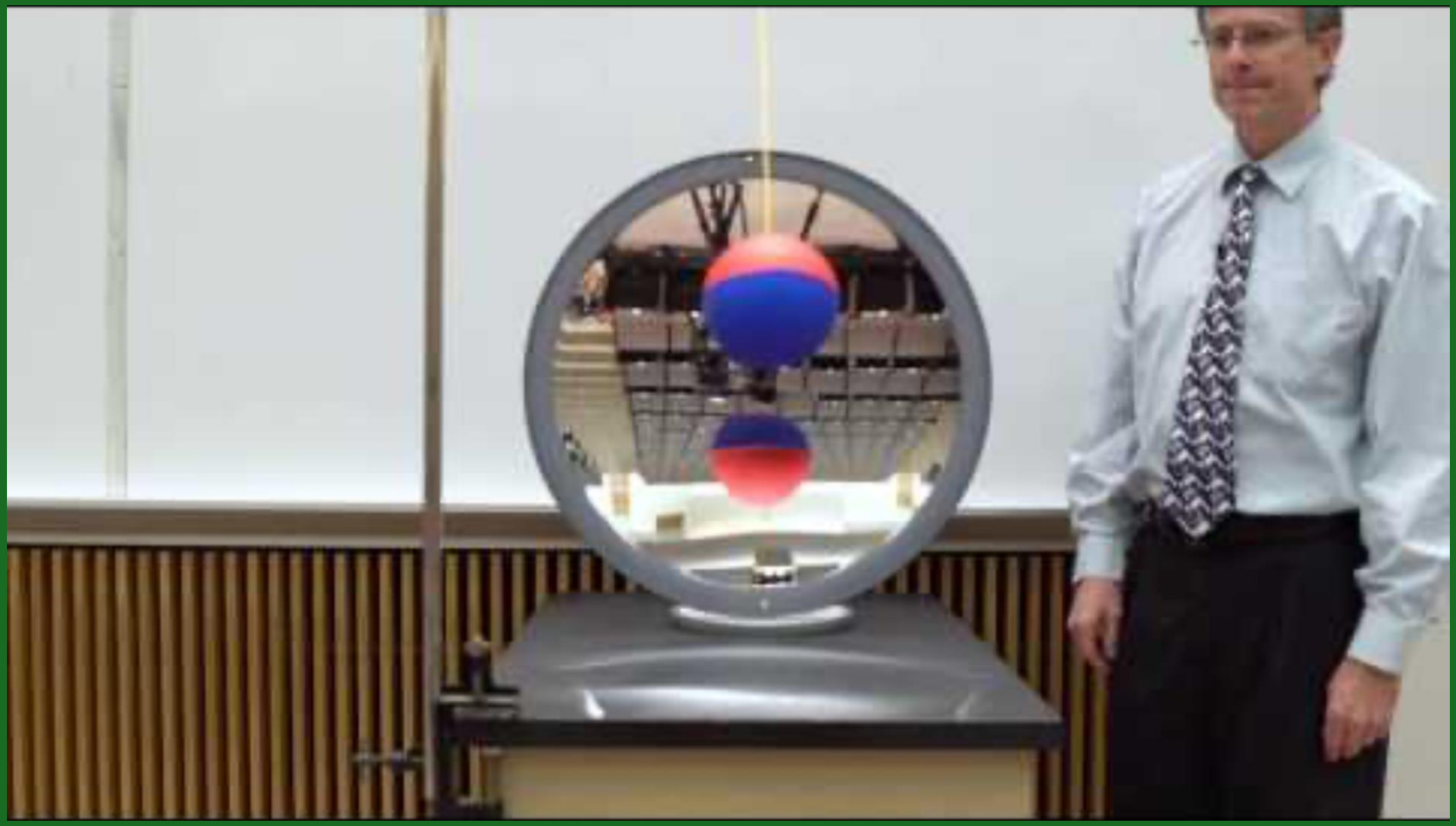


Curved Mirror

- Convex mirror (that curves outward): Virtual image is always smaller and closer to the mirror than the object.
- Concave mirror (that curves inward): Virtual image of a **close object** is larger and farther away than the object.



Concave Mirrors



Conceptual Question

Light reflecting from a smooth surface undergoes a change in

A. frequency.

B. speed.

C. wavelength.

D. None of the above is correct.

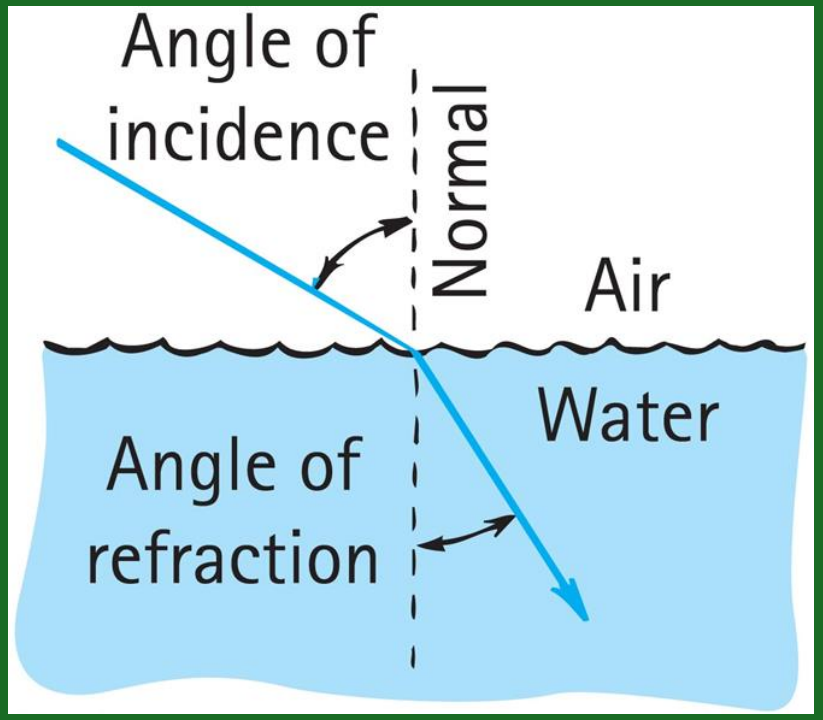
Conceptual Question

If you wish to take a picture of your image while standing 5 m in front of a plane mirror, for what distance should you set your camera to provide the sharpest focus?

A. 5 m
B. 10 m
C. Any distance will work
D. Not enough information

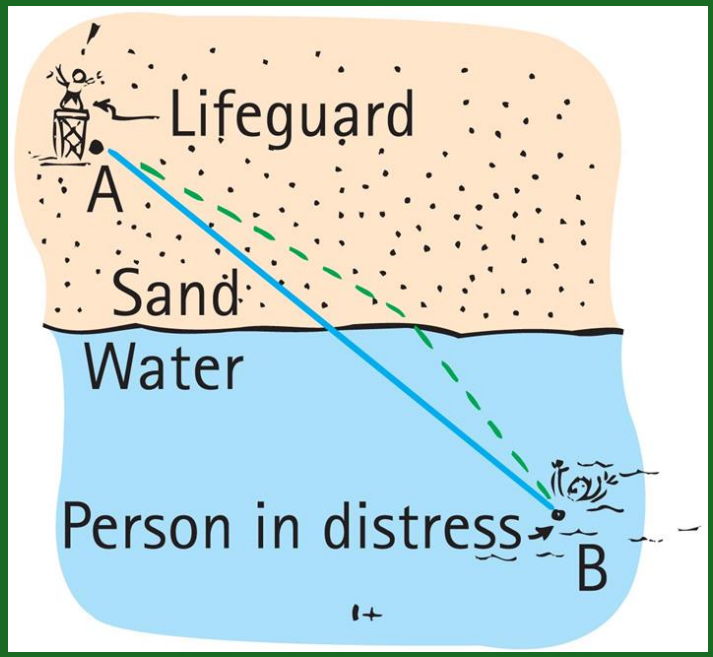
Refraction

Refraction is the bending of light rays after passing obliquely from one medium to another.



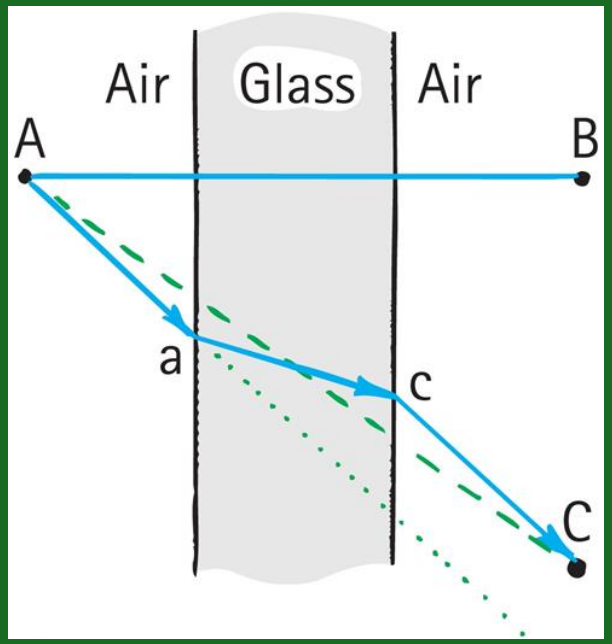
Quickest Path for Refraction

- Refraction occurs to minimize the time taken for light to travel from A to B, using a different light speed in different media.
- Similarly: To save someone from drowning, the quickest path would not be the straight line—it would be the dashed path shown. Presumably, the lifeguard moves quicker on the sand than in the water.



Refraction by Glass

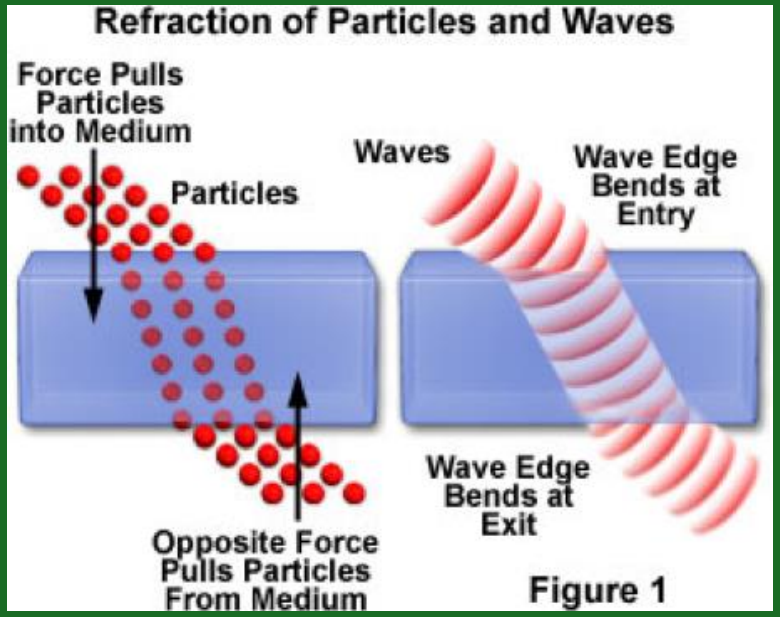
- Light travels slower in glass than in air, so it bends when passing from *a* to *c*, to minimize the time it spends in the glass.





Newton’s Corpuscular Model

- Newton’s corpuscular model proposed that light consists of tiny particles that speed up when entering a denser medium, like glass, due to an attractive force.
- This increase in speed causes light to bend toward the normal, explaining refraction. Later experiments showed light slows down in denser media, contradicting Newton’s model.



Conceptual Question

Refracted light that bends toward the normal is light that has

A. slowed down.

B. sped up.

C. nearly been absorbed.

D. diffracted.

Conceptual Question

Refracted light that bends away from the normal is light that has

A. slowed down.

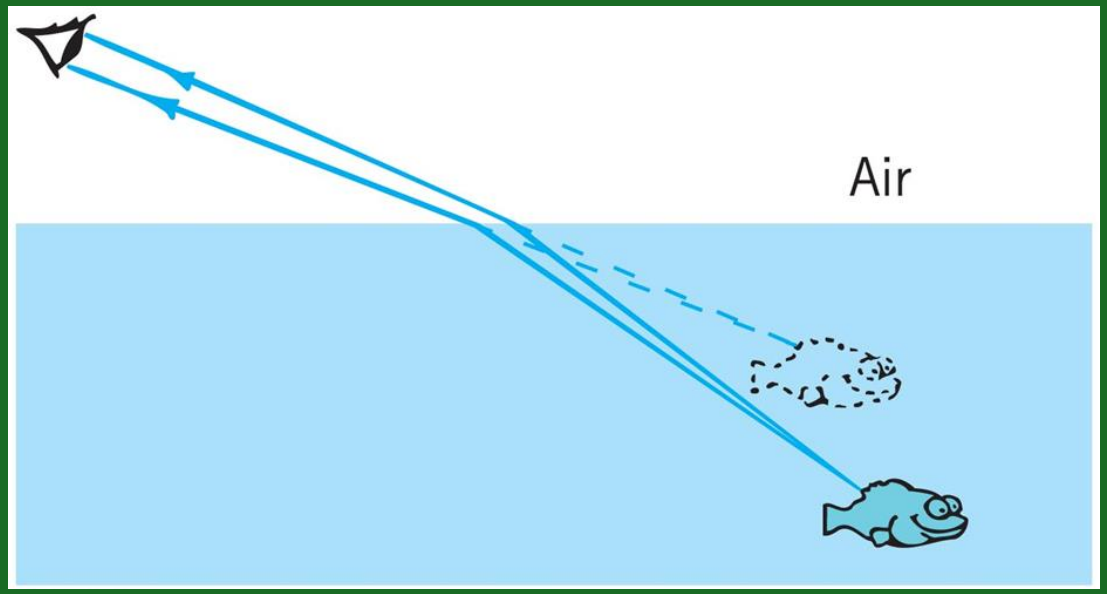
B. sped up.

C. nearly been absorbed.

D. diffracted.

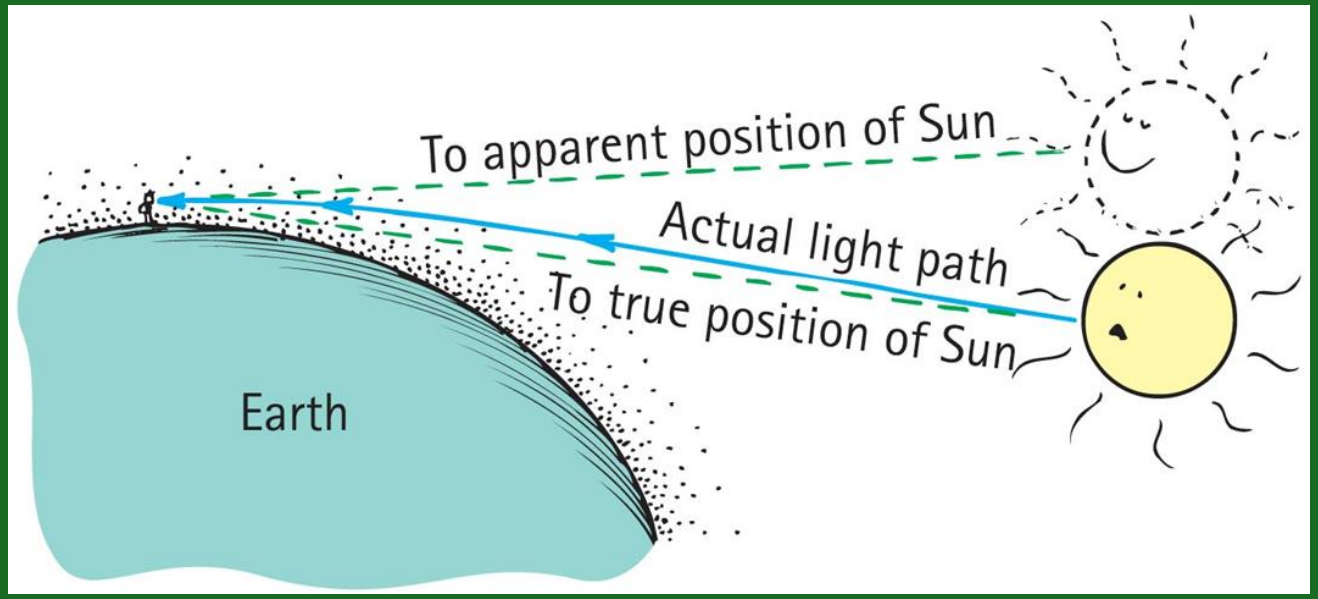
Illusions Caused by Refraction

- Objects submerged in water appear closer to the surface.



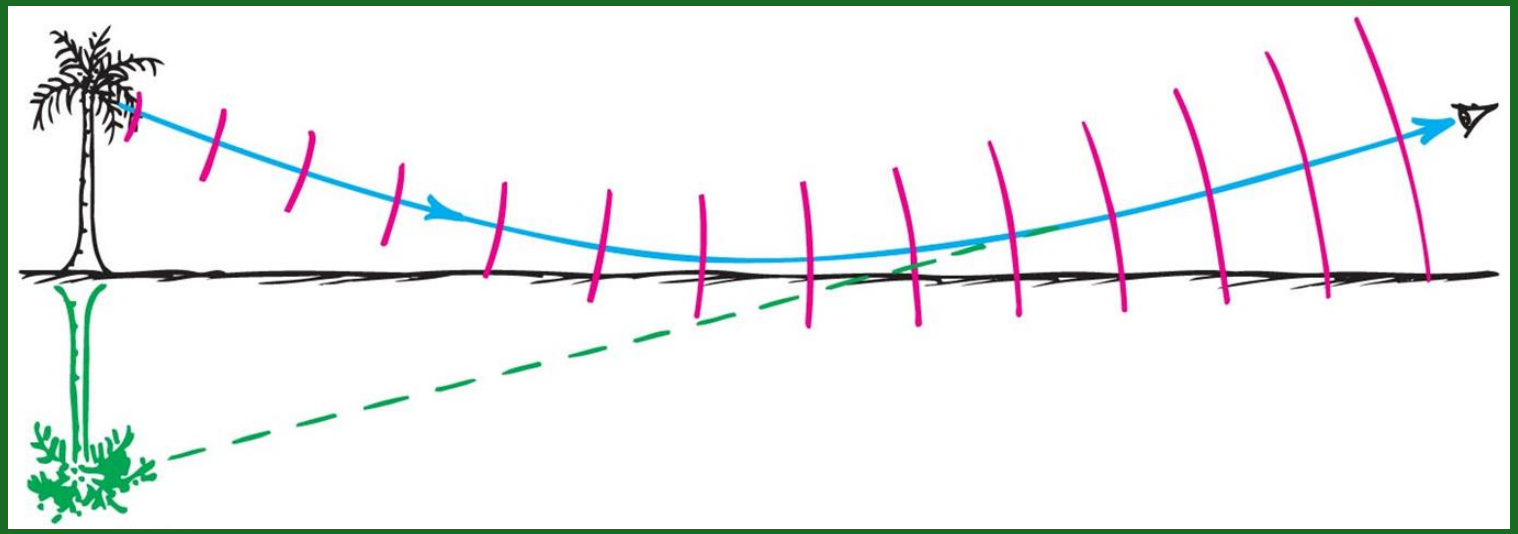
Illusions Caused by Refraction

- Objects such as the Sun seen through air are displaced because of atmospheric refraction.



Illusions Caused by Refraction

- Atmospheric refraction is the cause of mirages.



Index of Refraction

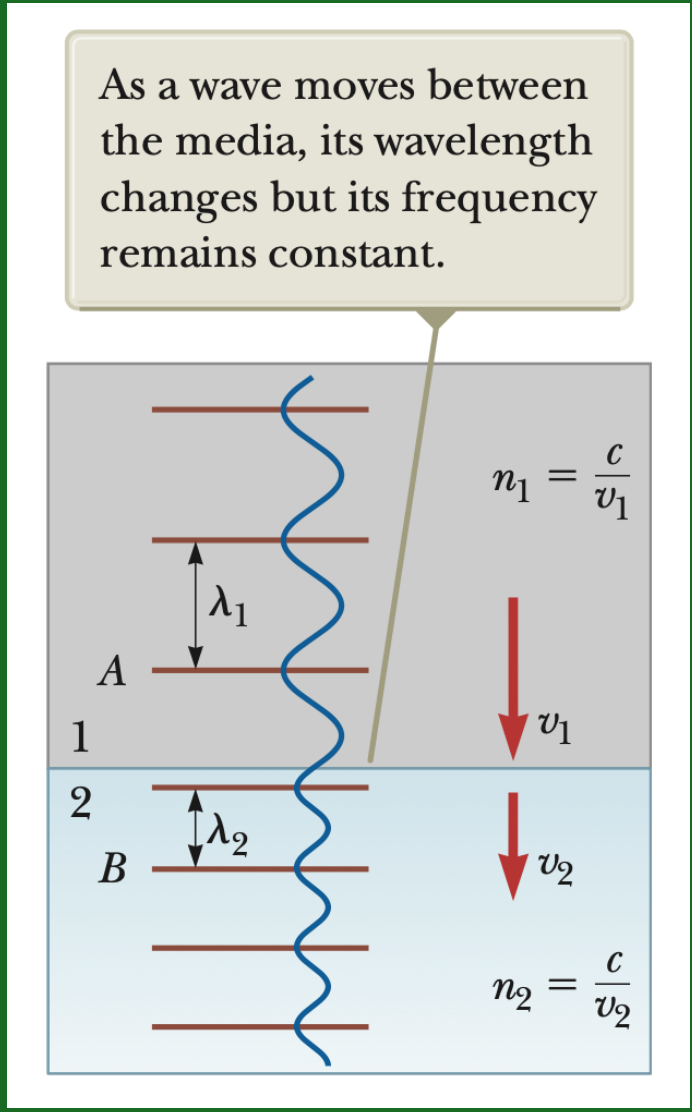
- The refractive index (index of refraction) of a material
  - is the ratio of speed of light in vacuum to the speed of light in the material.
  - determines the extent of bending of refracted light rays.
  - Medium with a high index means high bending effect and greatest slowing of light.

$$n = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$$

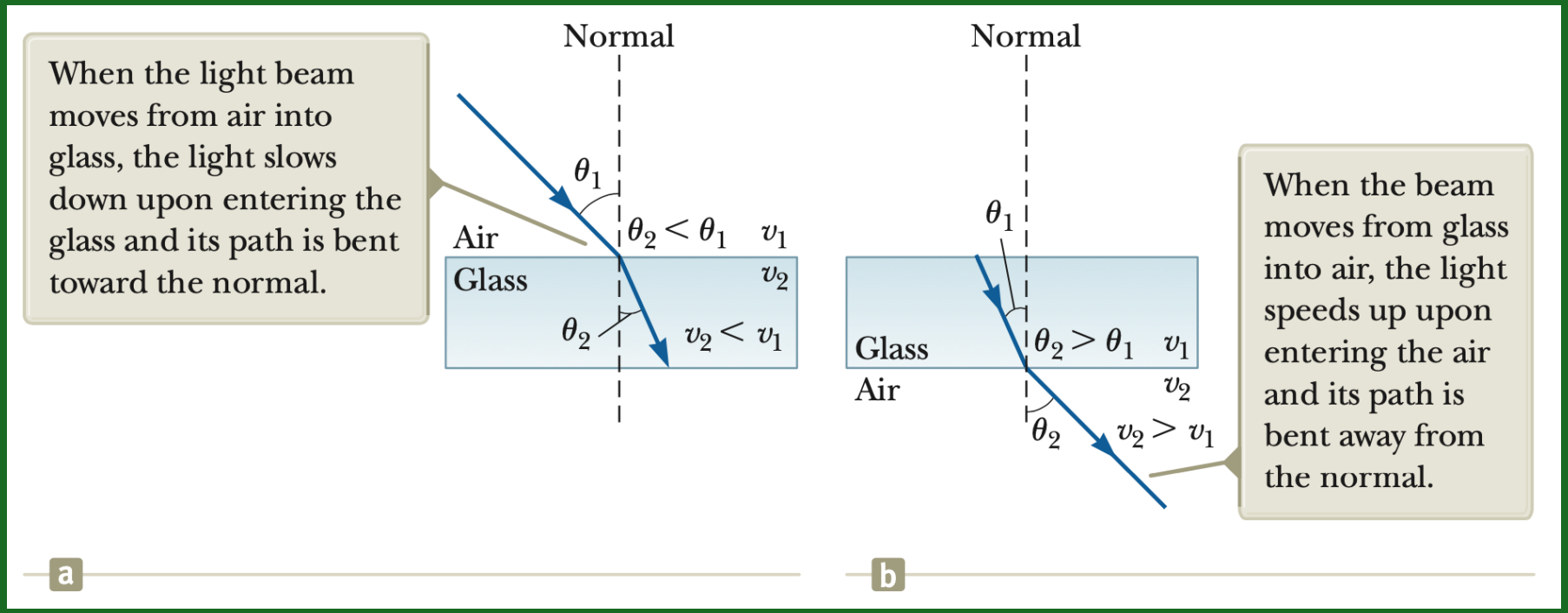
Refracted Wavelength

- A monochromatic light wave has a single wavelength/frequency, so $v = \lambda f$.
- When light enters a material, the speed changes but the frequency stays the same, so the wavelength must change.

$$\frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2} = \frac{c/n_1}{c/n_2} = \frac{n_2}{n_1}$$



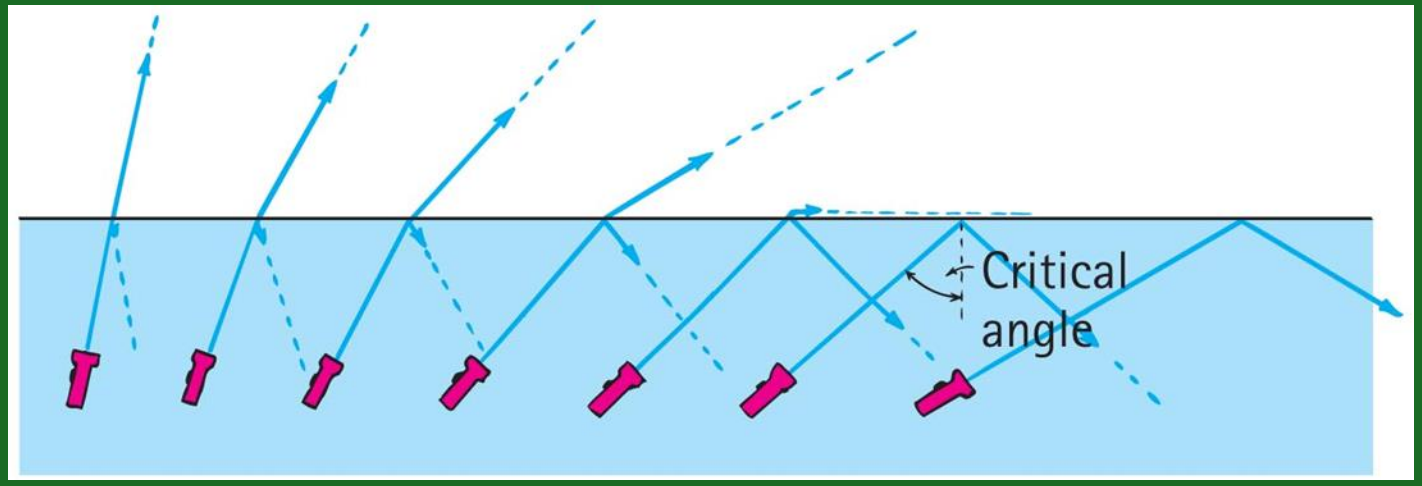
Law of Refraction (Snell's Law)



$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Total Internal Reflection

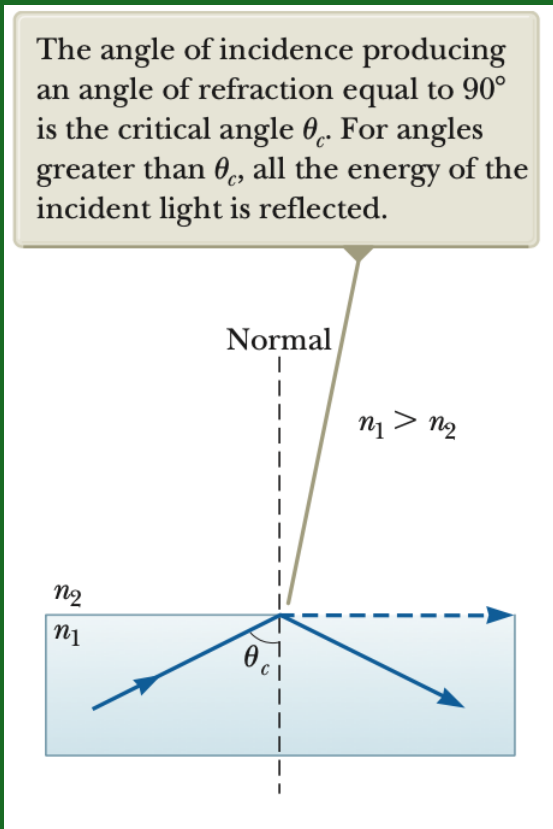
- Total internal reflection is total reflection of light traveling within a high-index medium that strikes the boundary of another low-index medium at an angle at, or greater than, a critical angle.



Critical Angle

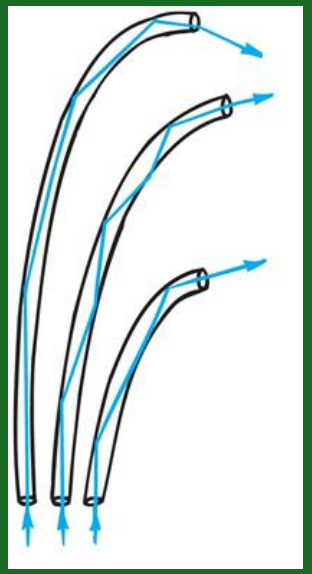
- Critical angle for light rays beneath the air/medium interface
  - minimum angle at which a light beam does not emerge into the air above the surface
  - varies for different materials

$$\sin\theta_c = \frac{n_2}{n_1} \quad (\text{for } n_1 > n_2)$$



Optical Fibers

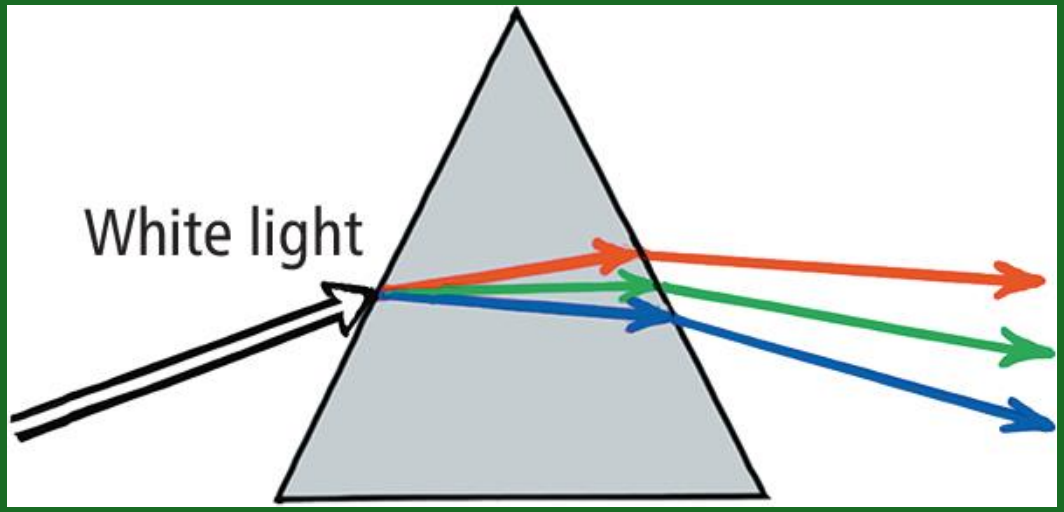
- Optical fibers are “light pipes”
  - Thin, flexible rods of special glass or transparent plastic.
  - Light from one end of the fiber is total internally reflected to the other end, resulting in nearly the same brightness of light emerging as entering
- Used in
  - illuminating instrument displays
  - concentrating light in dental procedures
  - viewing of inaccessible regions of organs and other devices communications





Dispersion

- Separation of light into colors arranged by frequency
- Dispersion is ultimately a *resonance effect;* some frequencies of light move slower than others because they interact more with the material.



Conceptual Question

Compared to the speed of green light passing through a prism, red light passing through the same prism

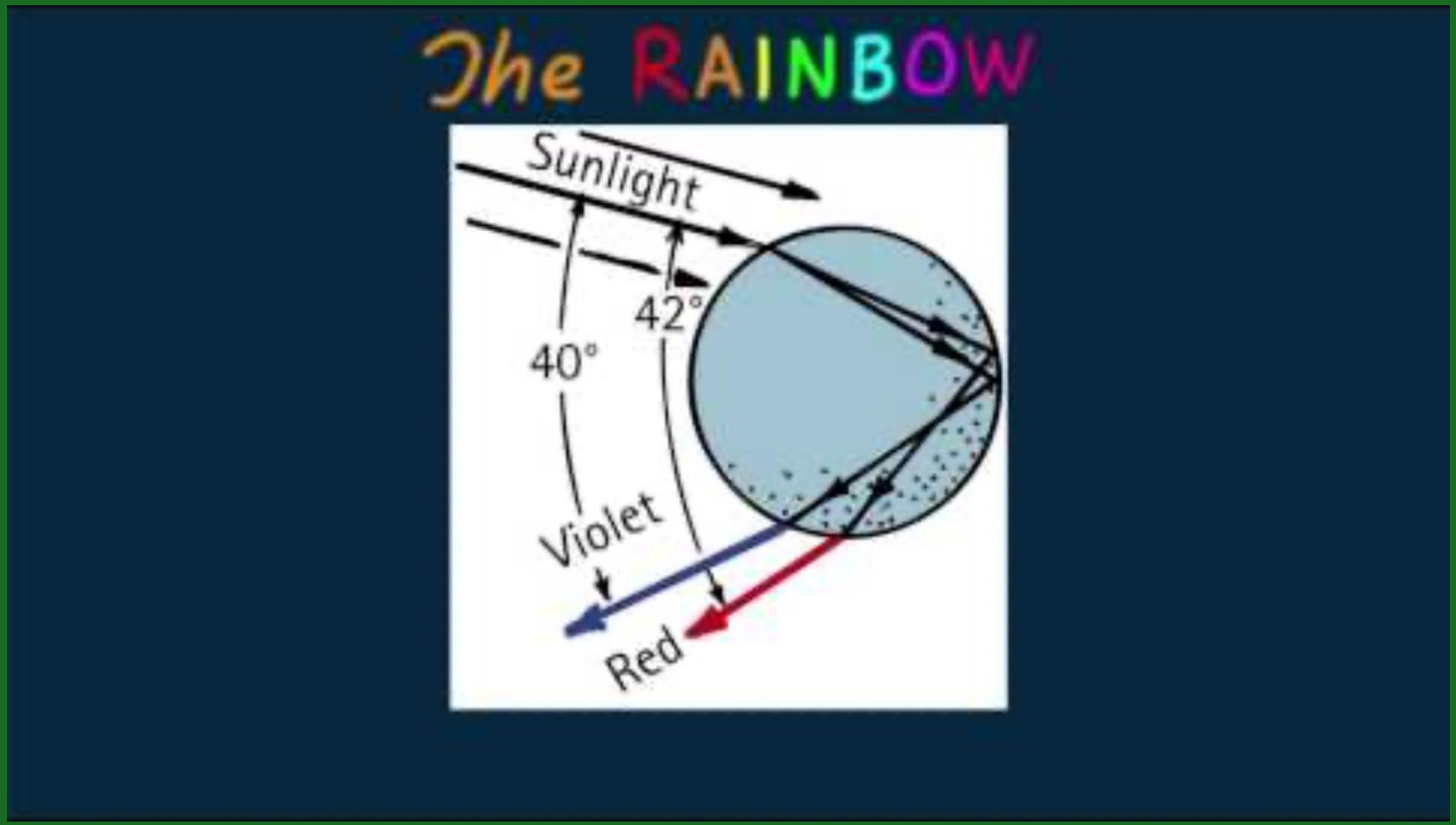
A. Moves faster

B. Moves slower

C. Moves at the same speed

D. Need more information

The Physics of Rainbows



Conceptual Question

Compared with the primary rainbow, the secondary bow

A. is dimmer.

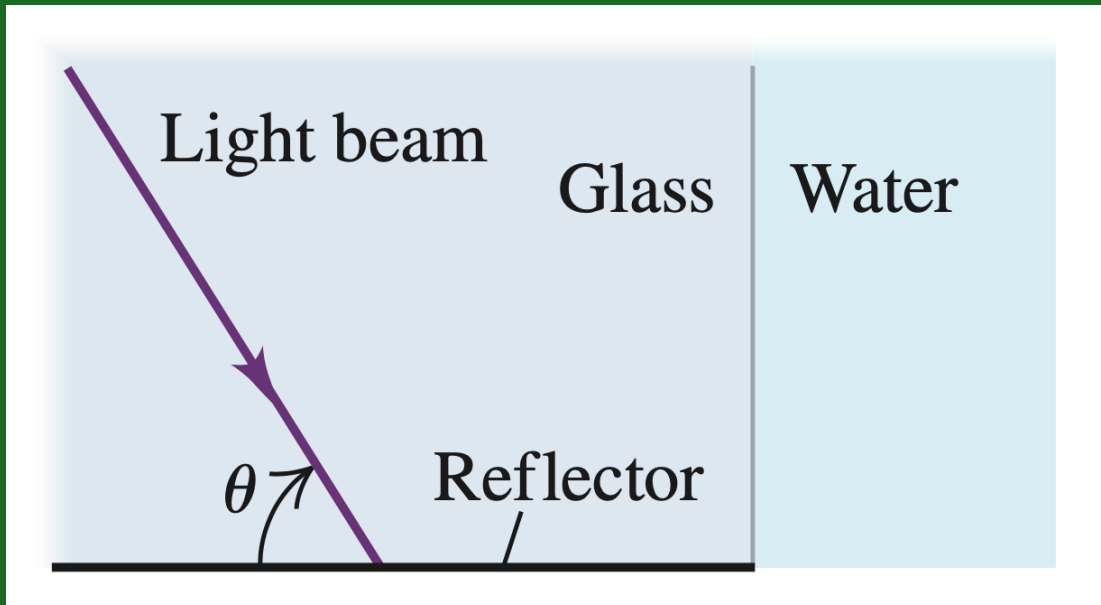
B. has colors reversed.

C. is caused by two internal reflections.

D. All of the above are correct.

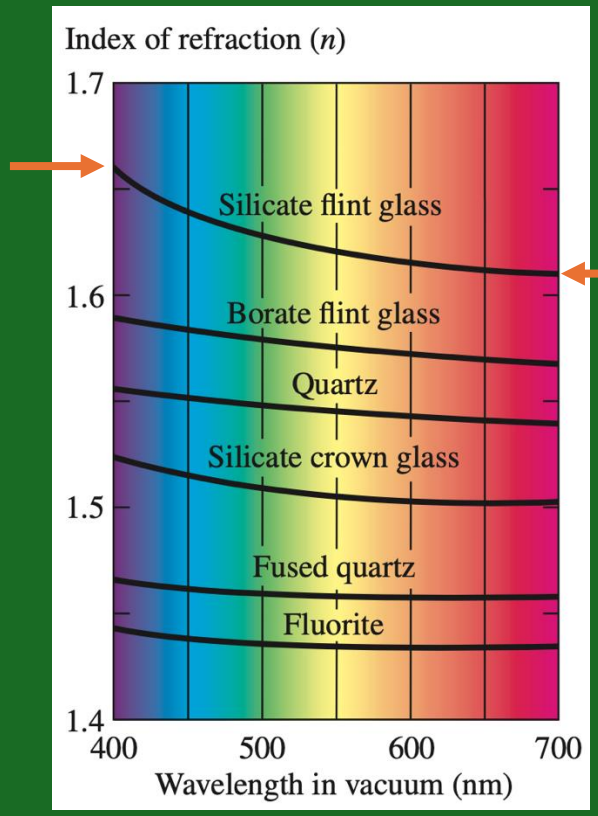
Example

The figure shows a rectangular glass block that has a metal reflector on one face and water on an adjoining face. A light beam strikes the reflector as shown. You gradually increase the angle θ of the light beam. If $\theta \geq 59.2°$, no light enters the water. What is the speed of light in this glass?



Example

A narrow beam of white light strikes one face of a slab of silicate flint glass. The light is traveling parallel to the two adjoining faces, as shown in the figure below. For the transmitted light inside the glass, through what angle $\Delta\theta$ is the portion of the visible spectrum between 400 nm and 700 nm dispersed? Use the information in the figure on the right.



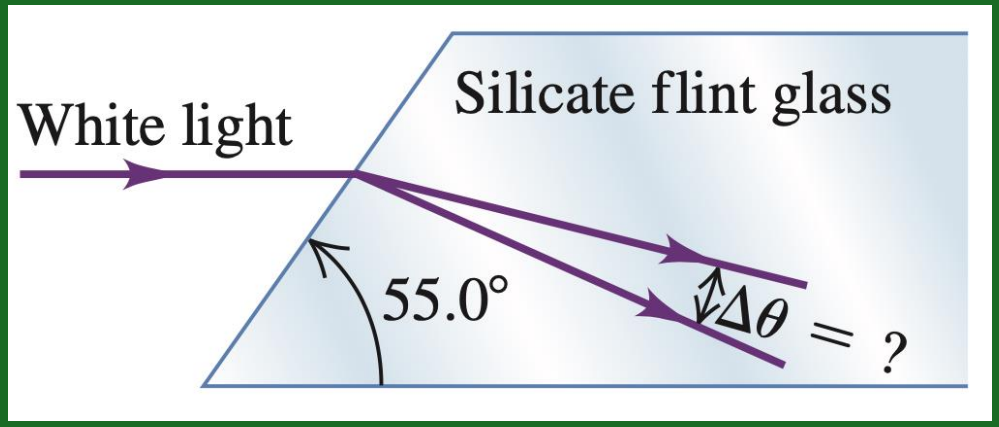
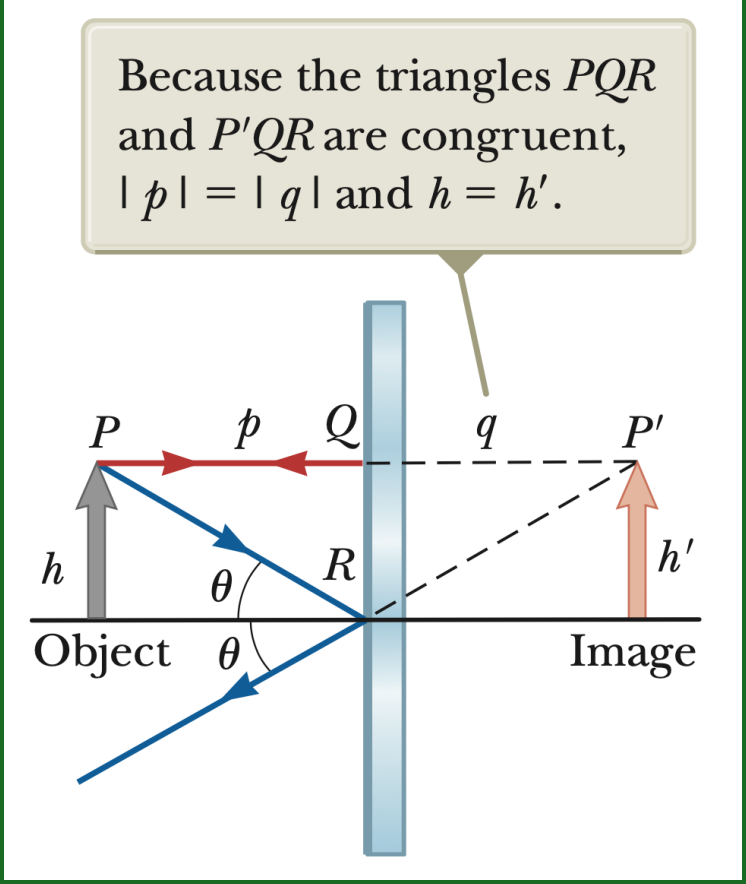


Image Formation

- An image is a pattern of light rays emerging from an optical system that carry information about an object's shape and position.
- Geometrically, we locate the image at the point or region from which these rays actually diverge (for a real image) or appear to diverge (for a virtual image).
- The image's position and size depend on how mirrors, lenses, or other optical elements reflect or refract the light.

Flat Mirrors

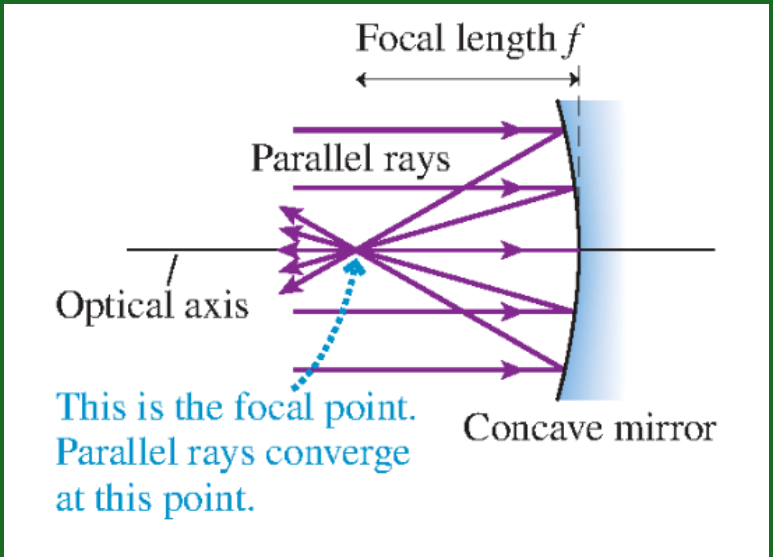


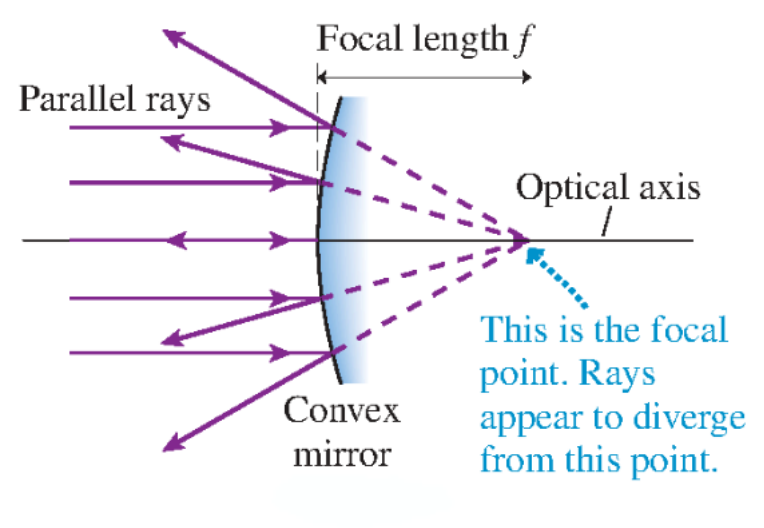
Lateral magnification

$$M = \frac{\text{image height}}{\text{object height}} = \frac{h'}{h}$$

$M = +1$ for a flat mirror

Spherical Mirrors





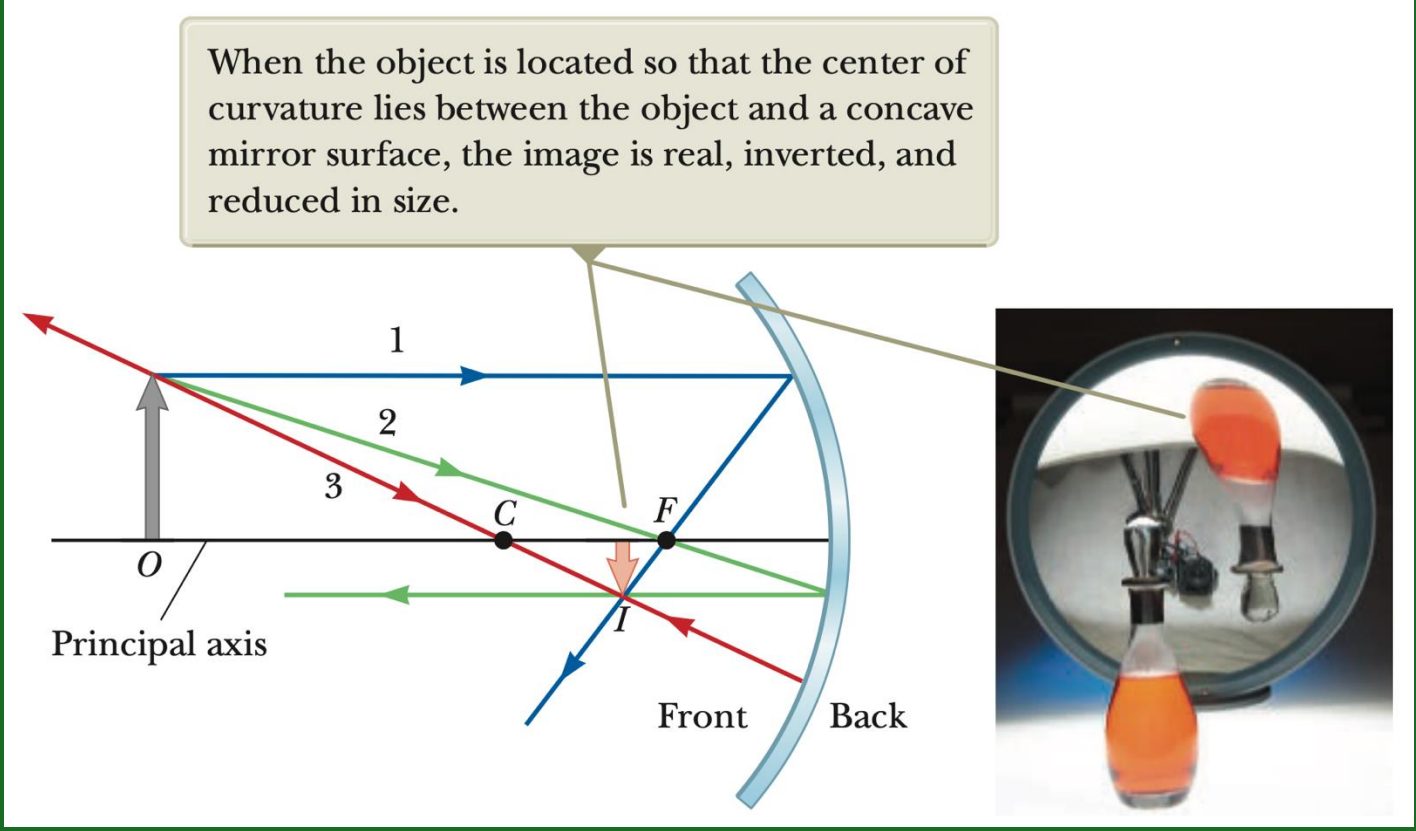
For paraxial rays (those that are parallel to the optical axis), the focal length of a spherical mirror is half its radius of curvature.

$$f = \frac{R}{2}$$

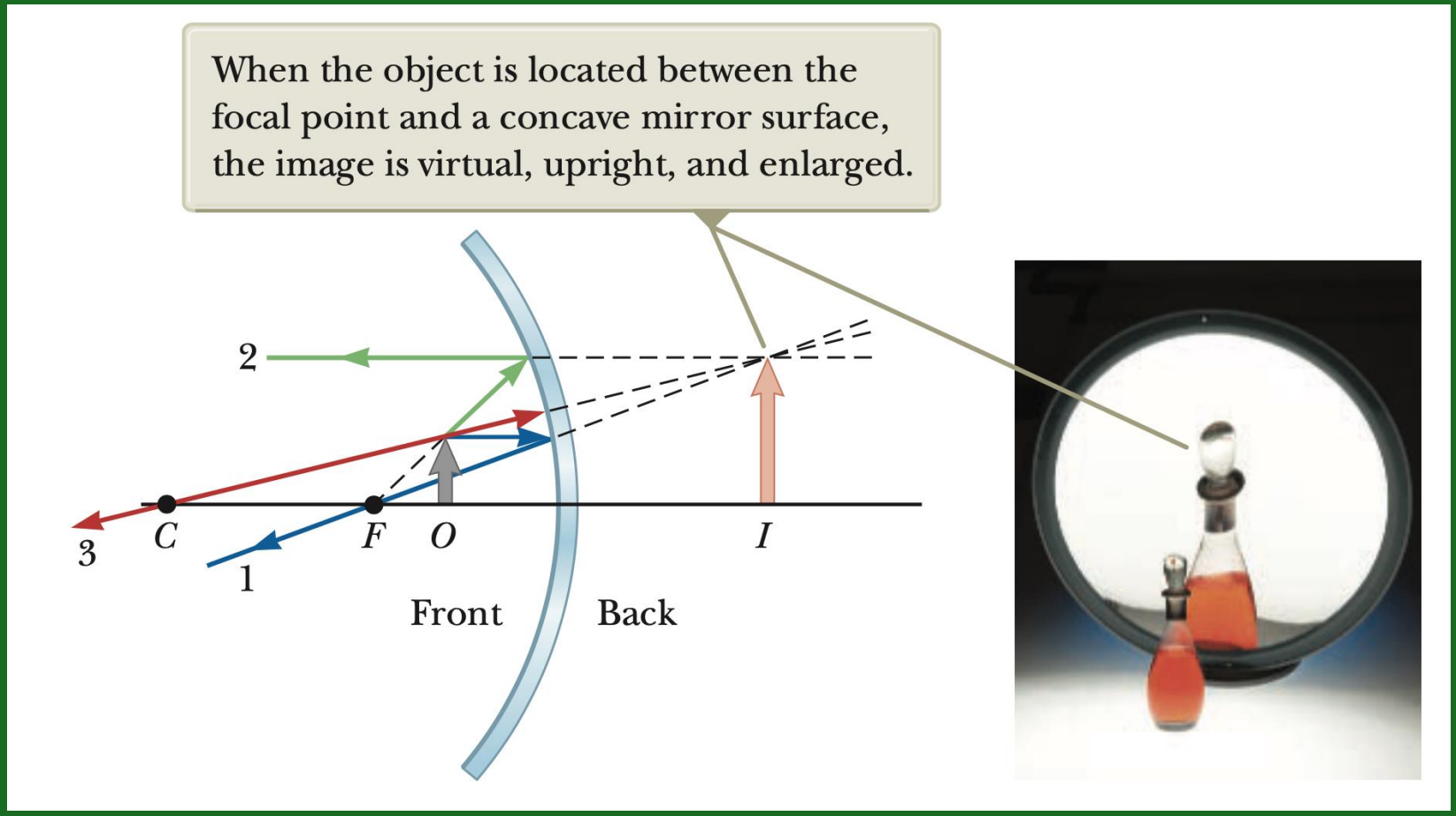
Ray Tracing for Spherical Mirrors

1. Center the mirror on the optical axis and indicate the focal point at distance f from the mirror's surface.
2. Represent the object with an upright arrow drawn from the axis at distance p from the mirror's surface.
3. Draw three special rays from the tip of the arrow. All reflections occur at the mirror plane.
    a. Ray #1 is parallel to the axis and reflects through (concave) or away from (convex) the focal point.
    b. Ray #2 passes through (concave) or away from (convex) the focal point reflects parallel to the axis.
    c. Ray #3 passes through the center of the sphere and reflects back on itself.
    d. Alternatively, you could choose Ray #4 that strikes the center of the mirror and reflects at an equal angle on the opposite side of the optical axis.
4. Extend the rays until they converge at the image point.

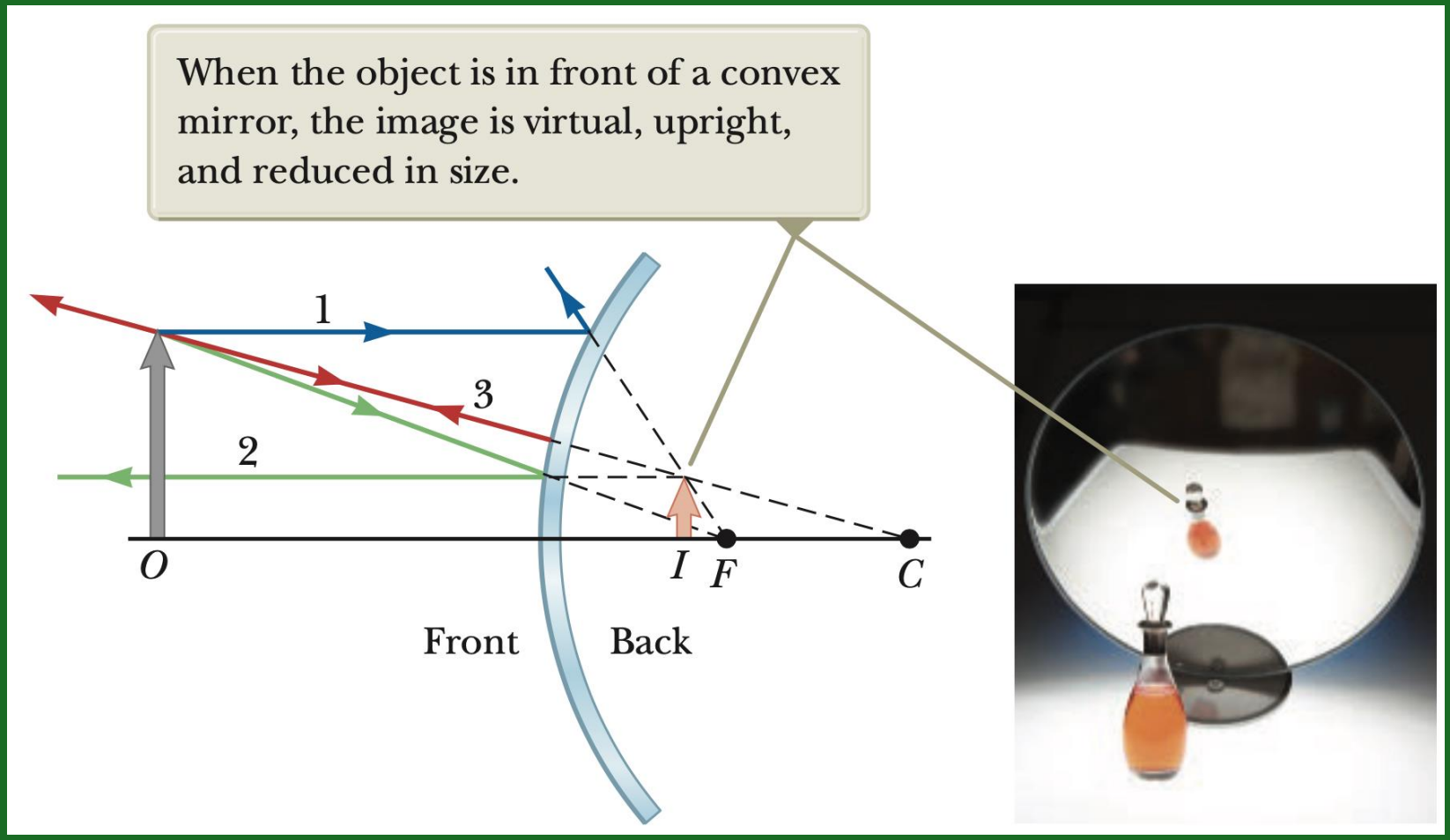
Concave Mirror ($p > f$)



Concave Mirror ($p < f$)



Convex Mirror



Mirror Equation & Sign Conventions

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

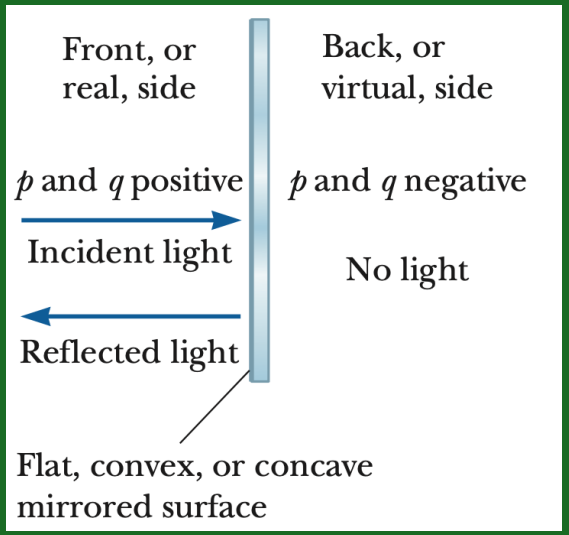


TABLE 35.1 Sign Conventions for Mirrors

Quantity	Positive When . . .	Negative When . . .
Object location (p)	object is in front of mirror (real object).	object is in back of mirror (virtual object).
Image location (q)	image is in front of mirror (real image).	image is in back of mirror (virtual image).
Image height (h')	image is upright.	image is inverted.
Focal length (f) and radius (R)	mirror is concave.	mirror is convex.
Magnification (M)	image is upright.	image is inverted.

Conceptual Question

A concave mirror of focal length f produces an image of the moon. Where is the image located?

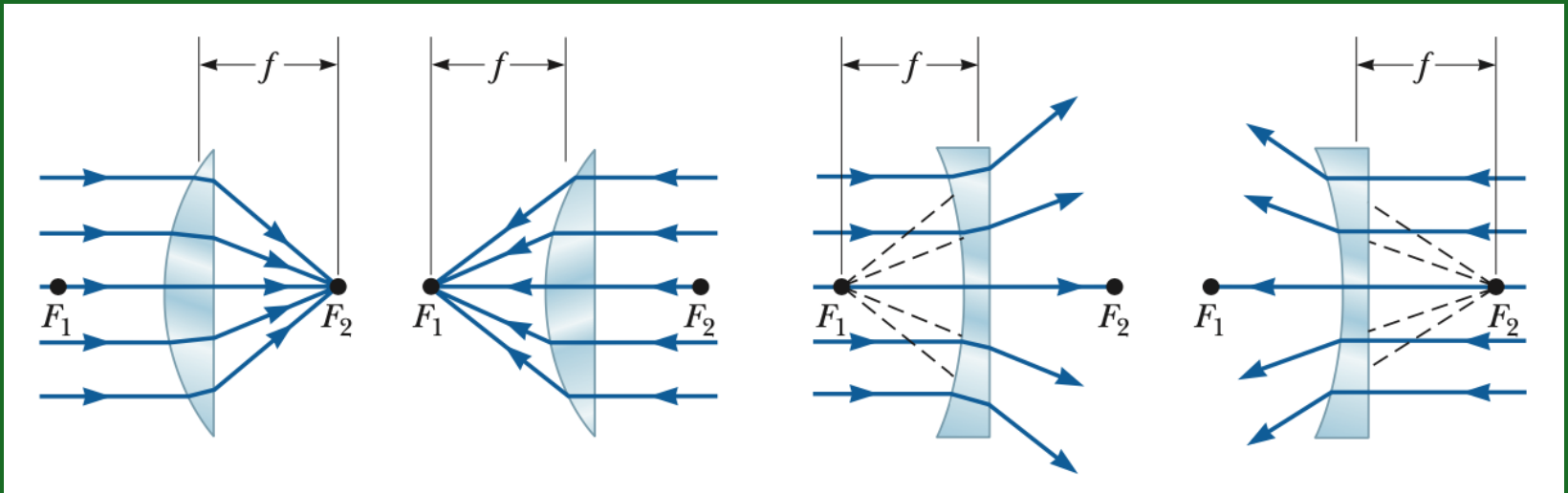
A. At the moon's surface.

B. Almost exactly a distance f behind the mirror.

C. Almost exactly a distance f in front of the mirror.

D. At a distance behind the mirror equal to the distance of the moon in front of the mirror.

Converging & Diverging Lenses



Converging Lens

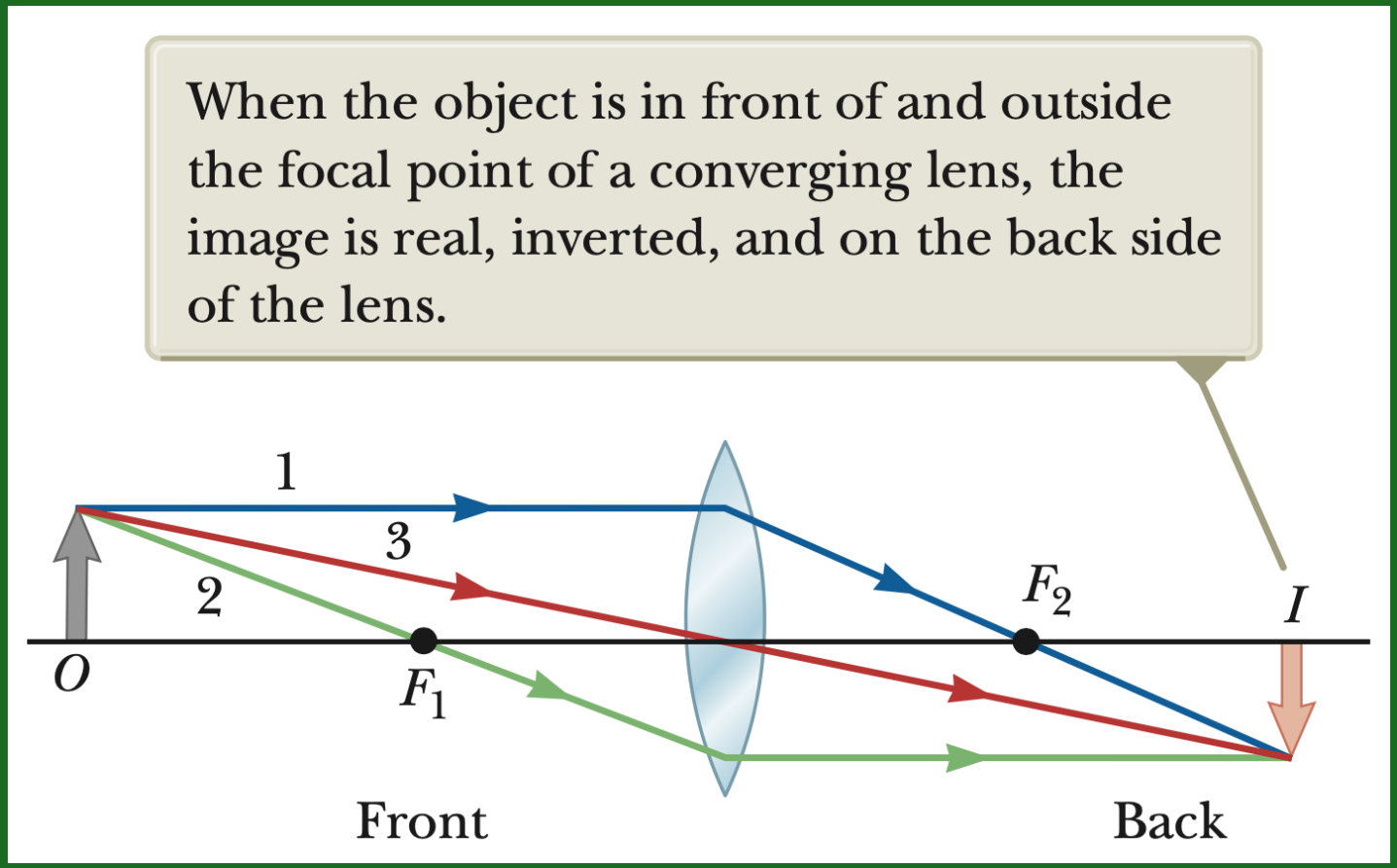
Diverging Lens

Ray Tracing for Thin Lenses

1. Ray #1 is parallel to the axis and refracts through the focal point.
2. Ray #2 passes through the focal point (or as if it comes from the focal point) and refracts parallel to the axis.
3. Ray #3 passes through the center of the lens and continues in a straight line.

Extend the rays until they converge to find the image point.

Converging Lens ($p > f$)

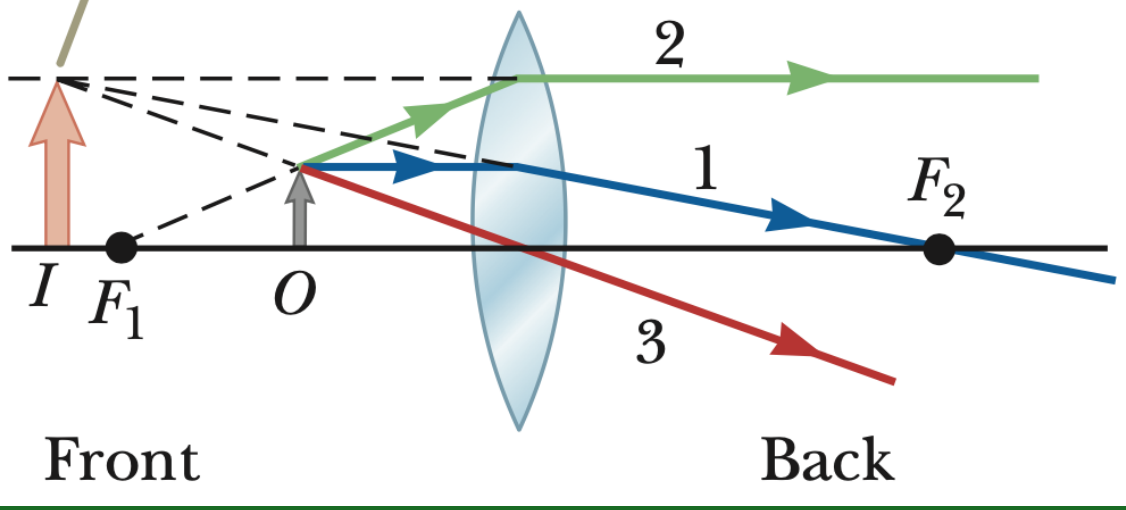


Converging Lens ($p > f$)



Converging Lens ($p < f$)

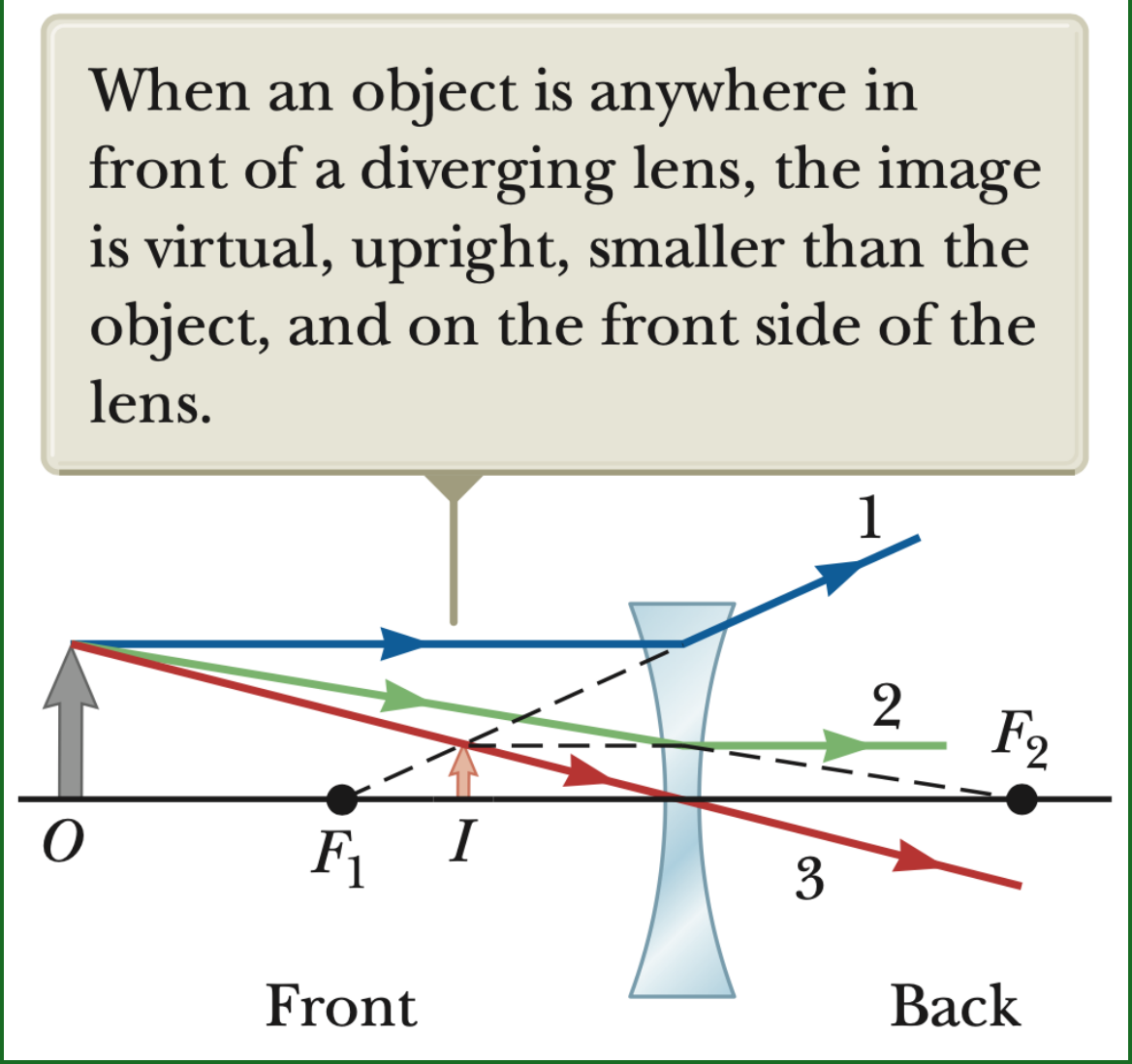
When the object is between the focal point and a converging lens, the image is virtual, upright, larger than the object, and on the front side of the lens.



Converging Lens ($p < f$)



Diverging Lens



Diverging Lens



Thin Lens Equation & Sign Conventions

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$$

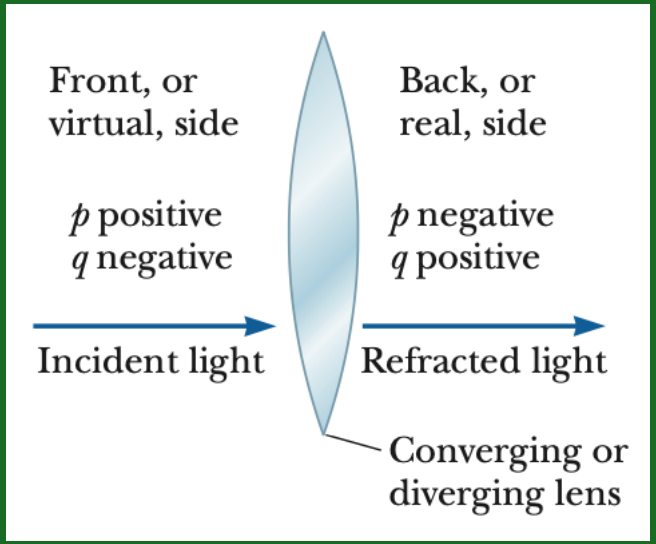


TABLE 35.3 Sign Conventions for Thin Lenses

Quantity	Positive When . . .	Negative When . . .
Object location (p)	object is in front of lens (real object).	object is in back of lens (virtual object).
Image location (q)	image is in back of lens (real image).	image is in front of lens (virtual image).
Image height (h')	image is upright.	image is inverted.
R_1 and R_2	center of curvature is in back of lens.	center of curvature is in front of lens.
Focal length (f)	a converging lens.	a diverging lens.

Conceptual Question

A lens produces a sharply focused, inverted image on a screen. What will you see on the screen if the lens is removed?

A. The image will be inverted and blurry.
B. The image will be upright and blurry.
C. The image will be upright and sharp.
D. The image will be upright but otherwise unchanged.
E. There will be no image at all.